CHAPTER XIII

ON RELATIONS

To be is to be related. (266) CASSIUS J. KEYSER

Science, in other words, is a system of relations. (417) H. POINCARÉ

Asymmetrical relations are involved in all series—in space and time, greater and less, whole and part, and many others of the most important characteristics of the actual world. All these aspects, therefore, the logic which reduces everything to subjects and predicates is compelled to condemn as error and mere appearance. (453) BERTRAND RUSSELL

My own investigations in this field, extending over some fifteen years, together with the facts already at hand, as I see them, have forced me to the conclusion that the organic individual is fundamentally . . . a system of relations between a physical substratum or structure and chemical reactions. (90) CHARLES M. CHILD

The thalamus, which in the lower vertebrates deprived of the cortex ensures the general reactions of the organism and the elementary mental functions, possesses an affective excitability in relation with the profound biological tendencies of the organism; among the higher mammals, indeed, it seems to preserve this rôle of affective regulation, whose importance in the behaviour of the organism and mental life is so often misunderstood. (411) HENRI PIÉRON

. . . organic impressions ('interoceptive' sensibility) appear in all cases to arrive at the cortex only when translated by the thalamus, with its own affective elaboration. (411) HENRI PIÉRON

Nevertheless, the consuming hunger of the uncritical mind for what it imagines to be certainty or finality impels it to feast upon shadows in the prevailing famine of substance. (22) E. T. BELL

In the foregoing chapters I made use of an expression, 'the organism-as-a-whole', which is employed continually in biology, psychiatry, and other branches of science. This expression is a restricted form of the general structural principle of non-elementalism. This expression implies that an organism is *not* a mere algebraic *sum* of its parts, but is *more* than that, and must be treated as an integrated whole. It was mentioned that the non-additivity and the 'more' than a mere 'sum' are complex problems which call for a new method of analysis. We have already seen that a simple analysis of the expression, 'Smith kicks Brown', involves a full-fledged structural metaphysics, or set of assumptions and terms which are taken on faith, since they cannot be defined, except circularly. In the present chapter, these subjects of great semantic importance will be developed further.

One of the fundamental structural defects and insufficiencies of the traditional *A*-system was that it had no place for 'relations', since it

assumed that everything could be expressed in a subject-predicate form. As we shall see, this is not true. Restriction to the subject-predicate form leaves out some of the most important structural means we have for representing this world and ourselves and has resulted in a general state of un-sanity. The explicit introduction of 'relations' is rather a recent innovation. A few words may be said about them, although the term 'relation' is one of the terms that we may accept as undefined, or that we may define in terms of multi-dimensional order.

Some relations, when they hold between A and B, hold also between B and A. Such relations are called *symmetrical.* For instance, the relation 'spouse'. If it holds between A and B, it holds also between B and A. If A is the spouse of B, B is the spouse of A. Terms like 'similarity' and 'dissimilarity' also designate relations of this kind. If A is similar or dissimilar to B, so is B similar or dissimilar to A. In general, a symmetrical relation is such that, if it holds between A and B, it also holds between B and A. In other words, the *order* in which we consider the relation of our entities is immaterial.

It is easy to see that not all relations are of such a character. For instance, in the relation 'A is the brother of B', B is not necessarily a brother of A, because B might be the sister of A. In general, relations which hold between A and B, but not necessarily between B and A, are called *non-symmetrical.* In these relations *order* becomes important. It is not a matter of indifference in what *order* we consider our entities.

If a relation is such that, if it holds between A and B, it *never* holds between B and A, it is called *asymmetrical.* Let us take, for instance, the relations 'father', 'mother', 'husband',. We readily see that if A is a father, or mother, or husband of B, B is *never* a father, or mother, or husband of A. The reversal of *order* is impossible in asymmetrical relations, and so any asymmetrical relation establishes a definite order.

Relations such as *before, after, greater, more, less, above, to the right, to the left, part,* and *whole,* and a great many others of the most important terms we have, are asymmetrical. The reader may easily verify this for himself. For instance, if A is *more* than B, B is *never* more than A,. We see at once that the troublesome little words, which are necessary to express *order* as 'before' and 'after'; terms of *evaluation,* such as 'more' and 'less'; and terms on which elementalism or non-elementalism depends, such as 'part' and 'whole', are in the list of asymmetrical relations.

Relations can be classified in another way, when three or more terms are considered. Some relations, called *transitive,* are such that, whenever they hold between A and B and also between B and C, they

hold between A and C. For example, if A is before, or after, or above, or more., than B, and B is before, or after, or above, or more., than C, then A is before, or after, or above, or more., than C.

It should be noted that all relations which give origin to series are transitive. But so are many others. In the above examples, the relations were transitive and asymmetrical, but there are numerous relations which are transitive and symmetrical. Among these are relations of equality, of being equally numerous,.

Relations which are not transitive are called non-transitive. For instance, dissimilarity is not transitive. If A is dissimilar to B, and B dissimilar to C, it does not follow that A is dissimilar to C.

Relations which, whenever they hold between A and B, *never* hold between A and C are called *intransitive*. 'Father', 'one inch longer', 'one year later'., are intransitive relations.

Relations are classified in several other ways; but, for our purpose, the above will be sufficient.

It is necessary now to compare the relational forms with the subject-predicate form of representation, which structurally underlies the traditional *A*-system and two-valued 'logic'. The structural question arises whether all relations can be reduced to the subject-predicate forms of language.

Symmetrical relations, which hold between B and A whenever they hold between A and B, seem plausibly expressed in the subject-predicate language. A symmetrical and transitive relation, such as that of 'equality', could be expressed as the possession of a common 'property'. A non-transitive relation, such as that of 'inequality', could also be considered as representing 'different properties'. But when we analyse *asymmetrical* relations, the situation becomes obviously different, and we find it a structural impossibility to give an adequate representation in terms of 'properties' and subject-predicates.

This fact has very serious semantic consequences, for we have already seen that some of the most important relations we know at present belong to the asymmetrical class. For example, the term 'greater' obviously differs from the term 'unequal', and 'father' from the term 'relative'. If two things are said to be unequal, this statement conveys that they differ in the magnitude of some 'property' without designating the greater. We could also say that they have different magnitudes, because inequality is a symmetrical relation; but if we were to say that a thing is unequal to another, or that the two have different magnitudes, when one of them was greater than the other, we simply should *not give an adequate account of the structural facts at hand.* If A is greater than

B, and we merely state that they are unequal or of different magnitudes, we *imply the possibility* that B is greater than A, *which is false to facts.* To give an adequate account, and to prevent *false implications,* there is no other way than to say which one is greater than the other. We see that it is impossible to give an *A adequate* account when asymmetrical relations are present. The possession of the 'same' 'property', or of different 'properties', are both *symmetrical relations* and seem covered by the subject-predicate form. But it is impossible to account adequately for asymmetrical relations in terms of 'properties'. In other words, we see that a language and 'logic' based upon subject-predicate structure may perhaps express symmetrical relations, but fail to express adequately asymmetrical relations, because both 'sameness' and difference of predicates are symmetrical.[1] Asymmetrical relations introduce a language of *new structure,* involving new *s.r.* Yet asymmetrical relations include many of the most important ones. They are involved in all *order,* all *series,* all *function,* in 'space', in 'time', in 'greater' and 'less', 'more' and 'less', 'whole' and 'part', 'infinity', 'space-time',. If we are restricted to the use of forms of representation unfitted for the expression of asymmetrical relations, ordinal, serial, functional, and structural problems could not be dealt with adequately. We should also have many insoluble semantic puzzles in connection with 'space', 'time', 'cause and effect', and many other relations in the world around us, and ourselves.

A very interesting structural and semantic fact should be noticed: that in symmetrical relations *order* is immaterial, in non-symmetrical relations it is important, and in asymmetrical relations *order* plays an all-important role and cannot be reversed. Order itself is expressed in terms of asymmetrical relations; as, for instance, 'before' or 'after', which apply to 'space', to 'time', 'space-time', 'structure'., and also to *all processes* and activities, the activities of the nervous system included. The asymmetrical relations 'greater', 'father'., imply ordering, while the 'unequal' (having different 'properties') or a 'relative'., do *not* imply ordering. If we consider subject-predicate forms as expressing a relation between the 'observer' and the 'observed', excluding humans, this last relation is also asymmetrical. Applying correct symbolism: if a leaf appears green to me, I certainly do not 'appear green' to the leaf! The last remark suggests that any *A* revision of the *A*-system is structurally impossible. To attempt a revision, we must begin with the formulation of a $\bar{A}$-system of different structure.

The above simple considerations have very far-reaching consequences, as without relations, and particularly without asymmetrical relations, we cannot have *order,* and without order, in the analysis of

processes, we are bound to introduce explicity or implicitly some objectively meaningless 'infinite velocities' of the propagation of the process. Thus, the 'infinite velocity' of light, which is known to be false to facts, is at the very foundation of the *N*-system. The equally false to facts silent assumption of the 'infinite velocity' of nerve currents underlies *A* animalistic 'psychology' and results in elementalism. This *el* 'psychology', until this day, vitiates all human concerns and even all science, the newer quantum theories not excluded.

General non-elementalism and, in particular, its restricted aspect, the 'organism-as-a-whole', implies the relation of the 'parts' to the 'whole', for which we need asymmetrical relations. In the statement '*more* than an algebraic sum', 'more' is also an asymmetrical relation. When we analysed the statement, 'Smith kicks Brown', we saw that the problems of 'space', 'time', 'infinity'., entered, the solution of which requires *serial* notions, which evade analysis without asymmetrical relations.

The solution of the problems of 'space' and 'time' are fundamental for a theory of sanity, as they are potent structural factors in all *s.r.* In the majority of 'mentally' ill, we find a disorientation as to 'space' and 'time'. Similar milder forms of disorientation appear in all forms of semantic disturbances, as they are disturbances of evaluation and meanings in the form of delusional 'absolute space' and 'absolute time'. These semantic disturbances can be eliminated only by considerations of multi-dimensional order, which are impossible without asymmetrical relations, and so could not have been accomplished in an *A*-system.

The problems of multi-dimensional order and asymmetrical relations are strictly interdependent and are the foundation of structure and so of human 'knowledge'; and they underlie the problems of human adjustment and sanity. Without going into details, I shall suggest some relational and ordinal aspects as found in the structure and function of the human nervous system and their bearing on semantic reactions and sanity. I shall also apply these considerations to the analysis of a historically very important delusional factor which has influenced, until now, the *s.r* of mankind away from sanity. I am dealing only with selected topics, important for my purpose, which, to the reader, may appear one-sided and unduly isolated. In fact, all issues involved are strictly interconnected in a circular way, and no verbal analysis of objective levels can ever be 'complete' or 'exhaustive', and this should be remembered. On the *A* silent assumption of the infinite velocity of nervous impulses, that the nervous impulses spread 'instantaneously', 'in no time' (to use an Alice-in-Wonderland expression), order was of no importance. But when we take into account the *finite* and known velocity

of nervous impulses, and the *serial,* chain structure of the nervous system, order becomes paramount. In such a serial structure, the problems of resistance, 'inhibitions', blockage, activation., become intelligible, so that some sane orientation is possible in this maze. It may be added that the intensity and the transformation of nervous impulses must somehow be connected with the paths they travel and are, therefore, problems to be spoken about in terms of order.

What has just been said may be illustrated by a rough and oversimplified hypothetical diagram. Fig. 1 shows how the normal (survival in man) impulse should travel. It should pass the thalamus, pass the sub-cortical layers, reach the cortex, and return. That the impulse is *altered* in passing this complicated chain is indicated in the diagram by the arbitrarily diminishing thickness of the line of the impulse.

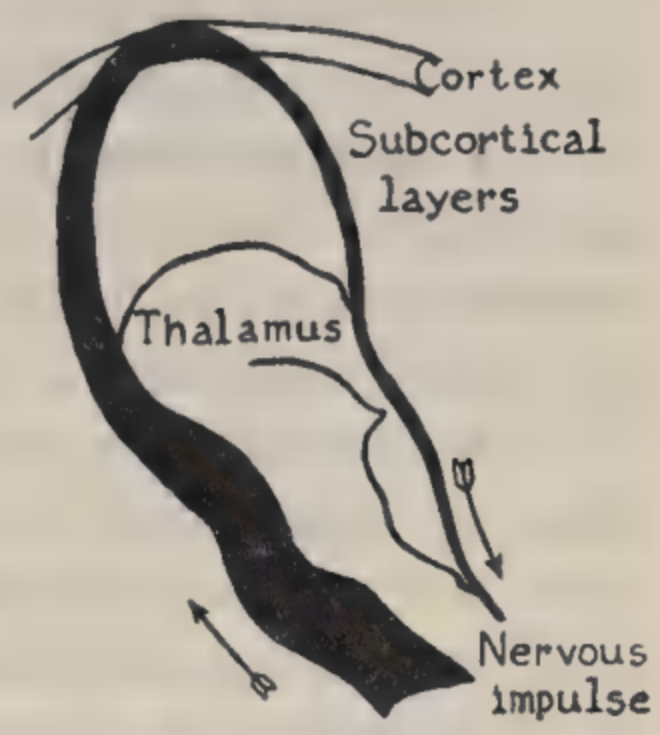

FIG. 1

Fig. 2 illustrates an hypothetical abnormal (non-survival in man) impulse. It emerges from the lower centres. For some reason or other, the main impulse is blocked semantically, or otherwise, and does not reach the cortex; only a weak impulse does. What should be expected in such a case? We should expect regression to the level of activities of organisms which have no cortex, or a cortex very little developed. But this could not be entirely true, as organisms without a cortex have a nervous system adequate for their lives, activities., in their environment, with survival values. But a higher organism with a cortex, no matter how rudimentary, has the other parts of the nervous structure quite different in function, and without the cortex they are *inadequate* for survival, as experience shows. We see that the *order* in which the impulses pass, or are deviated from their survival path, is paramount. A great many different reasons may produce such deviation, too many to list conveniently. A great many of them are known, in spite of the fact that, in general, we know very little about nerve mechanisms. Suffice to say, that we know, on colloidal grounds and from experience, that macroscopic or microscopic lesions, drugs, and *false doctrines* affecting

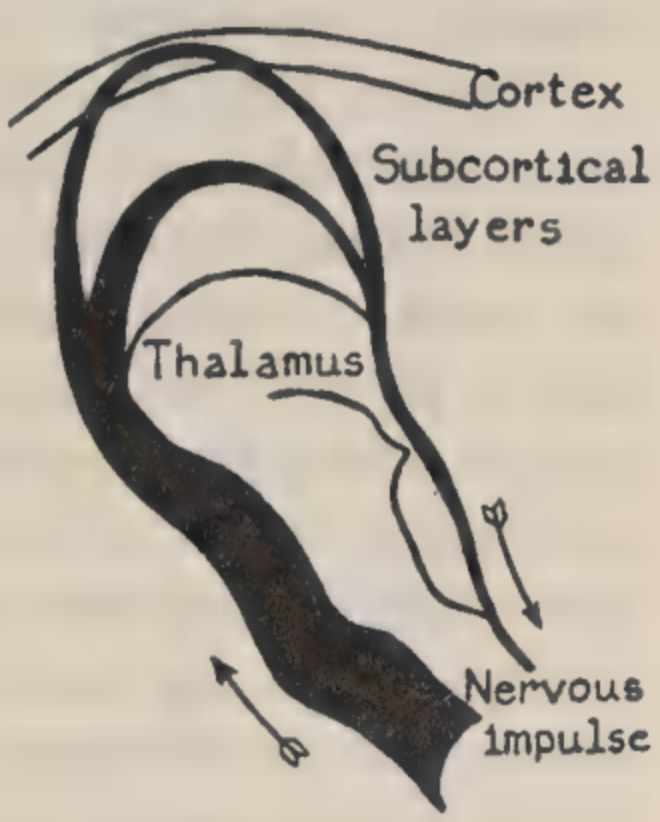

FIG. 2

the sub-microscopic levels, may often produce similar end-results. Here I use the term 'false doctrines' in the *non-el* sense, and, therefore, take into account affective and *evaluation*-components, which are usually disregarded when we speak about 'false doctrines'.

Here we must consider a problem of crucial, general human significance. It seems evident that *evaluation* in life, and particularly in human life, represents a most fundamental psycho-logical process underlying motivation and, in general, *s.r*, which shape our behaviour and result in collective structures which we may call 'stages of civilization'.

We may distinguish three periods of human development as characterized by their standards of evaluation:

1) The pre-human and primitive period of literal, general, and unrestricted identification. The semantics of this period could be formulated roughly as 'everything is everything else', which might be called one-valued semantics.

2) The infantile, or A period of partial or restricted identification, allowing symmetrical relations, to the exclusion of asymmetrical relations. Its semantics involve, among others, the 'law of identity'—'everything is identical with itself', its two-valued character being expressed by the postulate 'A is B or not B'.

3) The adult, or $\bar{A}$, or scientific period based on the complete elimination of identification, by means of asymmetrical and other relations, which establishes *structure* as the foundation of all 'knowledge'. Its semantics follow the ∞-valued semantics of probability and recognize 'equality', 'equivalence'., but no 'identity'.

Before analysing the above three periods separately, it must be stated that 'identity', defined as 'absolute sameness', necessitates 'absolute sameness' in *'all'* aspects, never to be found in this world, nor in our heads. Anything with which we deal on the objective levels represents a process, different all the 'time', no matter how slow or fast the process might be; therefore, a principle or a premise that 'everything is identical with itself' is *invariably false to facts.* From a structural point of view, it represents a foundation for a linguistic system non-similar in structure to the world or ourselves. All world pictures, speculations and *s.r* based on such premises must build for us delusional worlds, and an optimum adjustment to an *actual* world, so fundamentally different from our fancies, must, in principle, be impossible.

If we take even a symbolic expression $1 = 1$, 'absolute sameness' in *'all'* aspects is equally impossible, although we may use in this connection terms such as 'equal', 'equivalent',. 'Absolute sameness in all aspects' would necessitate an *identity* of different nervous systems which produce

and use these symbols, an *identity* of the different states of the nervous system of the person who wrote the above two symbols, an identity of the surfaces., of different parts of the paper, in the distribution of ink, and what not. To demand such impossible conditions is, of course, absurd, but it is equally absurd and very harmful for sanity and civilization to preserve until this day such delusional formulations as *standards of evaluation,* and then spend a lifetime of suffering and toil to evade the consequences. This may be comparable to the spending of many years in teaching and training our children that one and one *never* equal two, that twice two *never* equal four., and then they would have to spend a lifetime full of surprises and disappointments, if not tragedies, to learn, when they are about to die, that the above statements are always correct in mathematics and very often true in daily life, and finally acquire the sadly belated wisdom that they were taught false doctrines and trained in delusional *s.r* from the beginning.

If we revised these false doctrines, we would not twist the lives of younger generations to begin with. It seems that, for the sake of sanity, the term 'identity', symbolizing such a fundamental false structural doctrine, should be entirely eliminated from the vocabulary, but the term 'identification' should be retained in psychiatry as a label for extremely wide-spread delusional states which, at present, in a mild form, affect the majority of us.

If we investigate the standards of evaluation of animals, the experiments of Pavlov and his followers show that, after establishing a 'conditional reflex' (which means a physiological relating of a signal with food, for instance), the *physiological* effect of the signal on the nervous system of the animal is to produce secretions similar in quantity and quality to those the food produces. We can thus say that, from a physiological point of view, the animal organism *identifies* the signal with food. That represents the *animal standard of evaluation* at that given period. But even the animal nervous system is flexible enough to learn by experience that identification has no survival value, for, if, after the signal, food is repeatedly not forthcoming, he identifies again the signal with the absence of food. In more complex experiments, when both these identifications are interplayed, the result is a real physiological dilemma, culminating, usually, in a more or less profound nervous disturbance, corresponding to 'mental' ills in humans.

Identification represents a comparatively unflexible, rigid form of adaptation, of low degree conditionality, so to say, and, by neurological necessity, represents the processes of *animal* adaptation, inadequate for modern man. On human levels, it is found best exemplified in primitive

peoples and in cases of 'mentally' ill. In less severe cases of semantic disturbances, we also find identification of different degrees of intensity. The milder cases are usually considered as 'normal', which, in principle, is very harmful, because it establishes an animalistic, or primitive, standard of evaluation for 'normal'. 'Identity', as we have seen, is invariably false to facts; and so identification produces, and must produce, non-survival *s.r*, and, therefore, must be considered *pathological* for modern man.

That identification afflicts the majority of us today is also shown by experiments with conditional reflexes and the psychogalvanic experiments which show clearly that the majority of humans *identify* the symbol with actualities, and *secretions very often follow*. In other words, the reactions are of such a low order of conditionality as we find in animals and in primitive men. In principle, it makes no difference whether a sound (or word), or other signal (symbol) is identified with food or other actualities which are not symbols, and the secretions are produced by the adrenal glands, for instance, resulting in fear or anger, instead of by the salivary or sweat glands. In all such cases, *in experiments* with humans, the evaluation is false to facts, and the *physiological secretion* is uncalled for if the *evaluation would be appropriate* to the situation. In very few instances, the human experiments with conditional and psychogalvanic reflexes break down, in the sense that the signal-symbol is *not identified* with first order actualities, and so such an organism has no uncontrolled glandular secretions for signal-symbols *alone*. In a $\bar{A}$-system of evaluation, which involves on semantic levels the consciousness of abstracting, these exceptional persons (1933), with proper evaluation and controlled reactions, *prove the rule* for modern man. In other words, modern man, when he stops the pre-human and primitive identification, will have a much-increased and *conscious control* of his secretions, colloidal states of his nervous system . , and so of his reactions and behaviour. The above applies to all *s.r*, 'logical' processes included.

Identification is found in all known forms of 'mental' ills. A symbol, in any form, or any *s.r* may be identified in value with some fictitious 'reality' at a given date, resulting in macro-physiological (glandular, for instance) or micro-physiological (colloidal . ,) activities or disturbances which result in particular semantic states and behaviour. It is impossible to deny that 'mentally' ill have inappropriate standards of evaluation, and that identification appears always as an important factor in pathological evaluations. Experiments with 'mentally' ill show clearly that this evaluation can be altered or improved by different chemical

agencies which affect the colloids of the nervous system, by environmental changes . , and by *changing the standards of evaluation* which, at present, is usually called 'psychotherapy'. The analysis of the mechanism of evaluation leads, naturally, to a generalized and *simplified* method, which may have not only a therapeutic but also an important new *preventive value*.

Literal identification is found in all primitive peoples and accounts for their semantic states, reactions, their metaphysics, low development . , but it is impossible, for lack of space, to go into details here.

The *A* standard of evaluation departed from literal identification to some extent. We still preserve in our school books as the most fundamental 'law of thought'—the 'law of identity'—often expressed in the form 'everything is identical with itself', which, as we have seen, is invariably false to facts. We do not realize that, in a human world, we are dealing at most only with 'equality', 'equivalence'. , at a given place and date, or by definition, but never with 'identity', or 'absolute sameness', disregarding entirely space-time relations, involving 'all' the indefinitely many aspects which, through human ingenuity, we often manufacture at will. In an actual world of four-dimensional processes and the indefinitely many 'aspects' manufactured by ourselves, adjustment in principle is impossible, or, at best, only accidental, if we retain 'identity'. The *A* evaluation was based on symmetrical relations of 'identity' and also partial 'identity', expressed even in our political, economic . , doctrines and corresponding behaviour, the analysis of which would require a special volume to be written, I hope soon, by some one.

Under the pre-human and primitive standards of evaluation, science was not possible. Under the *A* standards the beginnings of science became possible, but if science had not departed from those standards, we would have had no modern science. Lately, when the persecution of science has increasingly relaxed (not in all countries in a similar degree) and scientists were allowed to develop their disciplines with much less fear of persecution, sometimes even encouraged and helped by public interest, scientists found that they invariably had to build their own vocabularies of a distinctly, although unrealized, $\bar{A}$ character. The chasm between human affairs and science became wider and wider. The reason for it was that, in life, even at present, we preserve *A* standards of evaluation, and science mainly depends on subtler $\bar{A}$ means involving asymmetrical relations which alone can give us *structure*. I will return repeatedly, later on, to the $\bar{A}$ re-evaluation of the *A* standards of values.

The $\bar{A}$ evaluation is based on asymmetrical and other relations. I shall not attempt to summarize it here because the problems are very

large and this whole volume is devoted to that subject. Here I shall mention, once more, that only with $\bar{A}$ standards of evaluation does a scientific treatment of man and his affairs become possible. A $\bar{A}$-system depends on a complete elimination of identification which affects beneficially all our *s.r*, as experience and experiments show.

It has been already emphasized that in the human child the nervous system is not physically finished at birth, and that for some years thereafter it is plastic. Hence, the 'environment'—which includes languages, doctrines, with their structure, all connected with evaluation-components —conditions the future functioning of the system. The way in which the nervous system works, the 'sanity', 'un-sanity' and 'insanity' of the individual depends to a large extent on how this plastic and sensitive apparatus is treated, particularly in childhood. Because of the *serial structure* of the nervous system, the language and doctrines supplied should be of the structure necessary for the adequate representation of serial structures and functions. With the old *A* means this could not be accomplished.

At this point, it will be well to introduce an important semantic subject, to which we shall return later; namely, the connection between the primitive subject-predicate language and identification. For example, the statement, 'the leaf is green', is taken to imply 'greenness', which, by its verbal structure, has the character of a 'substantive' and implies some sort of objective independence. It is *not* considered as an *asymmetrical relation* between the observer and the observed and, accordingly, tends toward an *additive* implication. 'Greenness' is thus objectified and *added* to the leaf in describing a 'green leaf'. The objectified 'greenness' leads to an anthropomorphic mythology, which, in turn, involves and develops the undifferentiated projecting mechanism so fundamental in semantic disturbances. The objectification is evaluated structurally as a 'real' situation, and this introduces the non-survival *reversed* order evaluation in which the use of the 'is' of identity, resulting in identification, is the main factor. The stronger the structural 'belief' in the 'truth' of the representation, or, in other words, the more we identify the higher order abstractions with the lower, which, in fact, are different, the more dangerous becomes the 'emotional' tension in the form of unjustified *evaluation,* which, ultimately, must involve *delusional* factors, no matter how slight, and result in semantic disturbances. *Ignorance,* involving strong faith in the erroneous structural belief, is dangerously akin to more developed symptoms of 'mental' illness called illusions, delusions, and hallucinations. We are mostly semantic victims of the primitive doctrines which underlie the *A* structure of our

language, and so we populate the world around us with semantic phantoms which add to our fears and worries, or which lead to abnormal cheerfulness, well known among some 'mentally' ill.

It should be realized that in the *A* system of evaluation many individuals profit in various ways by what amounts to distracting the attention of mankind from actual life problems, which make us forget or disregard actualities. They often supply us with phantom semantic structures, while they devote their attention to the control of actualities not seldom for their personal benefit. If one surveys the *A* situation impartially, one occasionally feels hopeless. But, no matter how we now conspire one against another, and thus, in the long run, against ourselves, the plain realization that the difficulty is found in the standards of evaluation, establishes the necessary preliminary step to the escape.

It is a well-known fact that, in a large proportion of 'mental' ills, we find a semantic flight from 'reality' (*m.o*) when their 'reality' becomes too hard to endure. It is not difficult to see that different mythologies, cults . , often supply such structural semantic 'flights from reality'; and that those who actually help, or who are professionally or otherwise engaged in producing and promulgating such semantic flights, help mankind to be un-sane, to deal with phantoms, to create dream states , . There is no longer any excuse in the old animalistic law of supply and demand —that, because there is a demand for such flights, they should be supplied. That argument is not held to apply to those who peddle drugs or wood alcohol. The flights from reality always have the earmarks of 'mental' illness. Very often such actively engaged individuals are themselves ill to the point of hallucinations; they often 'hear voices', 'see visions', 'speak tongues', . Very often other morbid symptoms occur which are similar to those shown by the 'mentally' ill of the usual hospital types. It is not generally realized that, although the patient suffers intensely, he usually shows marked resistance to any attempt to relieve him of his semantic affliction. Only *after* he is relieved by semantic re-education does the patient realize how very *unhappy he was*.

The situation is very serious. There is a powerful well-organized system, with enormous wealth behind it, based on *A* and pre-aristotelian standards of evaluation which keeps mankind in delusional semantic states. Its members do their best, better than they know, to keep mankind *un-sane* in flights from 'reality', instead of helping to revise the *A* standards of evaluation and to reorganize the horrible 'realities', *all of our own making,* into realities less painful. The comparatively few psychiatrists are naturally not a match for such vast numbers of well-

organized men and women who, in their blissful ignorance, work in the opposite direction; and all of us pay the price.

The activities of these individuals often promulgate something similar to the well-known 'induced insanity'. Quite often paranoid or paranoiac and, more rarely, hypomanic patients can influence their immediate companions to such an extent that they join in believing in their delusions and copy their *s.r*. Susceptible associates begin to develop similar delusions and hallucinations and to pass through episodes themselves, perfectly oblivious to contradictions with external *m.o* reality. There are many paranoiac-like semantic epidemics of this kind on record. It is instructive to visit some 'meetings' and watch the performer and the audience. The pathetic side of it is that these performers, themselves not realizing the harmfulness of the situation, often pretend, or genuinely believe, that they are helping mankind by preaching some metaphysical 'morals'. What they *actually* produce is a disorganization of the survival-working of the human nervous system, particularly if they train the structurally undeveloped nervous system of children to delusional evaluations and *s.r*, and, in general, make sanity and higher and effective ethical standards very difficult or impossible. It is positively known that *s.r* are inextricably connected with electrical currents, secretions of different glands., which, in turn, exert a powerful influence on colloidal structure and behaviour, and so condition our neurological and physiological development. There can be no doubt that imposing delusional *s.r* on the undeveloped child must result in at least colloidal injury, which later on facilitates arrested development or regression, and, in general, leads away from adjustment and sanity.

Lack of space and the essentially constructive aims of the present system do not allow me to analyse many fundamental interrelations in the development of man, but a brief list, worthy of analysis, may be suggested:

1) The relation between the pre-human reactions and the reactions of the primitive man, involving always some *copying* by mutants of the responses of the prevailing simpler organisms.

2) The interrelation between the reactions of the primitive man, his animism, anthropomorphism, his other *s.r* and the *structure* of his language and semantics.

3) The relation between the structure of primitive languages and the structure of the 'philosophical grammar' formulated by Aristotle, generally called 'logic'.

4) The relation between this grammar, the structure of language, and the further development of our structural metaphysics and *s.r*.

5) The influence the last conditions exerted on the structure of our institutions, doctrines, and the *s.r* related to them.

6) The relation between the 'copying animals in our nervous processes' and semantic blockages . , preventing an adult civilization, agreement, sanity, and other desirable human reactions.

This brief list suggests an enormous field for further research, but, even now, the formulation of a $\bar{A}$-system of evaluations makes a few points clearer.

An infant, be it primitive or modern, begins life with *s.r* of identity and confusion of orders of abstractions, natural to his age, yet false in principle, and structurally false to fact. At present, parents and teachers seldom check or counteract this tendency, mostly not realizing the importance of this semantic factor and its role in the future adjustment of the individual. In the rough, to a baby, his cry 'is' food. Words 'are' magic. This identification is structurally false to facts, but in *babyhood it mostly works*. To the infant, experience proves that the noises he makes, a cry or a word, have the objective value,—food. The semantic identity of the symbol and the un-speakable object level,—food,—has been established. This infantile attitude or *s.r* is carried on into grown-up life.

Under very simple conditions of primitive peoples, in spite of many difficulties, this attitude of identification is not always checked by experience, and experimenting is non-existent at this stage. If it is, then such checking of identification is 'explained' by some sort of demonology and 'good' or 'evil' 'spirits',. Delusional, from the modern point of view, *s.r* are compensated by mythologies, making the two sides of the semantic equation equivalent. This equating tendency is *inherent* in all human *s.r*. It expresses the instinctive 'feel' for the similarity of structure as the base of 'knowledge', and it ultimately finds its expression in mathematical equations. In all psycho-logical processes of 'understanding', we must have some standards of evaluation and 'equivalence'. On primitive levels, this is accomplished by literal identification and delusional mythology of the type, that a storm at sea is 'caused' by a violent quarrel between a 'god' and his 'wife'; or, in contemporaneous mythology, a draught, or fire, or death by lightning, is explained as 'punishment' for 'sins',. Semantic compensation is needed and produced. A similar semantic process produces scientific theories, but with different standards of evaluation. At present, scientific theories do not cover all semantic needs and urges of mankind, owing to the prevailing false to fact identification of different orders of abstractions. With the full consciousness of abstracting, which means proper evaluation or differentiation between orders of abstractions, science will then cover all our non-pathological semantic

needs, and different primitive mythologies will become unnecessary. A very harmful, primitive, delusional semantic factor of blockages would be eliminated.

The 'is' of identity plays a great havoc with our *s.r*, as any 'identity' is structurally false to fact. An infant does not know and cannot know that. In his life, the 'is' of identity plays an important semantic role, which, if not checked intelligently, becomes a pernicious semantic factor in his grown-up reactions, which preserve the infantile character and with which *adult* adjustment and semantic health is impossible. The infant begins to speak and again he is trained in the 'is' of identity. Symbols are identified with the un-speakable actions, events and objects under penalty of pain or even death. The magic of words begins its full sway. As a rule, parental, crude disciplining of the infant, particularly in former days, trained the *s.r* of the infant again in the delusional 'is' of identity. The results are semantically and structurally very far-reaching and are found to underlie modern mythologies, militarism, the prevailing economic and social systems, the control by fear (be it 'hell' or machine guns), illusory gold standards, hunger , .

Experience shows that such identification of symbols with the un-speakable levels works very well with animals. With man, it leads only to the misuse of the human nervous system, semantic disturbances of evaluation, and the prevailing unstable animalistic systems in practically all fields, resulting in the general chaos in human affairs.

It should be noticed that the 'is' of predication also expresses a sort of *partial identity,* leading to primitive anthropomorphism and general confusion of orders of abstractions. By an inherent necessity, our lives are lived on the un-speakable objective levels, which include not only ordinary objects but also actions and immediate feelings, symbols being only auxiliary means. The natural ordinal evaluation, which should be the foundation for healthy *s.r,* appears as the event-process level first, the object next in importance; the objective level first, the symbolic next in importance; the descriptive level first, the inferential level next in importance , . The semantic *identification* of these different levels not only abolishes the natural evaluation, but, in fact, reverses the natural order. Once this is realized, we see clearly that all statements about the objective level, which is made up of absolute individuals, are only *probable* in different degrees and can never be certain. The 'is' of identity underlies, also, the two-valued, too primitive, too restricted, and structurally fallacious *A* 'logic'.

The crucial semantic importance of asymmetrical relations becomes obvious when we consider that all *evaluation* and *non-el* meanings

depend ultimately on asymmetrical relations. In the technical fields, mathematics and the exact sciences; in the semi-scientific fields, economics, politics, sociology.; in the as yet non-scientific fields, 'ethics', 'happiness', 'adjustment'., represent ultimately different forms of *evaluation,* impossible to formulate adequately under aristotelianism.

Obviously, a $\bar{A}$-system based on proper semantic evaluation leading to non-pathological reactions, adjustments., must make relations and multi-dimensional order fundamental for sanity. The semantic connection between mathematical methods and all the other concerns of man becomes also necessary and obvious.

In mathematics, recently, the notion of equality needed a refinement and the notion of 'identity' has been introduced. The present analysis discloses that, although the refinement and the symbol may be retained, yet the name should be entirely abandoned, because it conceals a very semantically vicious confusion of orders of abstractions. If, by definition, we produce new terms, these new terms are of a higher order abstraction than the terms used in the definition, and so the *identification* of them as to the orders of abstractions is *physiologically* and structurally false to facts.

The problems discussed in the present chapter have been felt vaguely for more than two thousand years and found their first historical expression in the rift between Aristotle, the biologist, and Plato, the founder of mathematical philosophy. Mathematics is, in principle, $\bar{A}$, and so, in the study of mathematics, we can learn most about the principles of non-aristotelianism. In physics, only very recently, do we begin to eliminate the 'is' of identity and elementalism which resulted in the $\bar{N}$ systems. All sciences strive to become more mathematical and exact and so $\bar{A}$. In fact, all advances in science are due to the building of new $\bar{A}$ languages, usually called 'terminology'. We can go further and say, definitely, that, to have any science, we must make a $\bar{A}$ revision of the languages used. Similarly with 'man', either we decide to introduce into human affairs scientific evaluation, and so part company with the A and pre-aristotelian system of evaluation, or preserve A structure, and have no science of man, or science of sanity, but continue in the prevailing chaos.

CHAPTER XIV

ON THE NOTION OF INFINITY

The questions on which there is disagreement are not trivialities; they are the very roots of the whole vast tree of modern mathematics. (22)

E. T. BELL

The task of cleaning up mathematics and salvaging whatever can be saved from the wreckage of the past twenty years will probably be enough to occupy one generation. (22)

E. T. BELL

The intention of the Hilbert proof theory is to atone by an act performed once for all for the continual titanic offences which mathematics and all mathematicians have committed and will still commit against mind, against the principle of evidence; and this act consists of gaining the insight that mathematics, if it is not true, is at least consistent. Mathematics, as we saw, abounds in propositions that are not really significant judgments. (549)

HERMANN WEYL

An objectivated property is usually called a *set* in mathematics. (549)

HERMANN WEYL

If the objects are *indefinite* in number, that is to say if one is constantly exposed to seeing new and unforseen objects arise, it may happen that the appearance of a new object may require the classification to be modified, and thus it is we are exposed to antinomies. *There is no actual (given complete) infinity.* (417)

H. POINCARÉ

The structural notion of 'infinite', 'infinity', is of great semantic importance and lately has again become a subject of heated mathematical debates. My examination of this subject is from the point of view of a $\bar{A}$-system, general semantic, and a theory of sanity which completely eliminates identification. In Supplement III, I give a more detailed $\bar{A}$ analysis of the problem already anticipated by Brouwer, Weyl, Chwistek, and others. These problems are not yet solved, because mathematicians, in their orientations and arguments, still use *el, A* 'logic', 'psychology', and epistemology, which involve and depend on the 'is' of identity, making agreement impossible.

Mathematical infinity was first put on record by the Roman poet, Titus Lucretius, who, as far back as the first century B.C., wrote very beautifully about it in his *De Rerum Natura*.[1] As the author was a poet, and his work poetry, a few privileged literati had great pleasure in reading it; but this discovery, not being rigorously formulated, remained inoperative, and so practically worthless for mankind at large, for 2000 years. Only about fifty years ago, mathematical infinity was rediscovered by mathematicians, who formulated it rigorously, without poetry. Since then, mathematics has progressed with all other sciences in an unprece-

dented way. That this structural linguistic discovery was made so late is probably due to the usual blockage, the old *s.r*, old habits of 'thought', and prejudices.

In all arguments about infinity, from remote antiquity until Bolzano (1781-1848), Dedekind (1831-1916), and Cantor (1845-1918), there was a peculiar maxim involved. All arguments against infinity involved a certain structural assumption, which, at first inspection, seemed to be true and 'self-evident', and yet, if carried through, would be quite destructive to all mathematics existing at that date. Arguments favorable to infinity did not involve these tragic consequences. Quite naturally, mathematicians, and particularly Cantor, began to investigate this peculiar maxim and the *s.r* which were playing havoc. The structural assumption in question is that 'if a collection is part of another, the one which is a part must have fewer terms than the one of which it is a part'. This *s.r* was deeply rooted, and even found a scholarly formulation in Euclid's wording in one of his axioms: 'The whole is greater than any of its parts'. This axiom, although it is not an exact equivalent of the maxim stated above, by loose reasoning, which was usual in the older days, could be said to imply the troublesome maxim. It is not difficult to see that the *E* axiom, as well as our troublesome maxim, expresses a structural generalization taken from experience which applies only to *finite* processes, arrays , . Indeed, both can be taken as a definition of finite processes, arrays , . It does *not* follow, however, that the one definition and structure must be true of infinite processes, arrays , . As a matter of fact, the break-down of this maxim gives us the precise definition of mathematical infinity. A process of generating arrays . , is called infinite when it contains, as parts, other processes, arrays . , which have 'as many' terms as the first process, array , .

The term 'infinite' means a process which does not end or stop, and it is usually symbolized by ∞. The term may be applied, also, to an array of terms or other entities, the production of which does not end or stop. Thus we may speak of the infinite process of generating numbers because every positive integer, no matter how great, has a successor; we can also speak of infinite divisibility because the numerical technique gives us means to accomplish that. The term 'infinite' is used here as an adjective describing the characteristics of a *process*, but should never be used as a noun, as this leads to self-contradictions. The term 'infinity', as a noun, is used here only as an abbreviation for the phrase 'infinite process of generating numbers', . If used in any other way than as an abbreviation for the full phrase, the term is meaningless in science (not in psychopathology) and should never be used. The above semantic

restrictions are not arbitrary or purely etymological, but they follow the rejection of the 'is' of identity of a $\bar{A}$-system.

Before we can apply the term 'infinite' to physical processes, we must first theoretically elucidate this term to the utmost, and only then find out by experiment whether or not we can discover physical processes to which such a term can be applied. Fortunately, we have at our disposal a *semantic process* of generating number*s* which, by common experience, by definition, and by the numerical technique, is such that every number has a successor. Similarly, our semantic processes are capable by common experience, by definition, and by the numerical technique to divide a finite whole indefinitely. Thus, if we do not identify external physical objective processes with internal semantic processes, but differentiate between them and apply correct symbolism, we can see our way clear. If we *stop* this semantic process of generating number*s* at any stage, then we deal with a finite number, no matter how great; yet the process remains, by common experience, by definition, and by the numerical technique, such that it can proceed indefinitely. In the $\bar{A}$ sense, 'infinite', as applied to processes, means as much as 'indefinite'. We should notice that the semantic *process* of *generating* number*s* should not be identified with a *selection* of *a* definite number, which, by necessity, is finite, no matter how great. The identification of the semantic process of generating number*s* with a definite number; the identification of the semantic process of infinite divisibility of finites in the direction of the small with the generating of number*s* in the direction of the great; and the identification of semantic internal processes with external physical processes . , are found at the foundation of the whole present mathematical scandal, which divides the mathematical world into two hostile camps.

The process of infinite divisibility is closely connected with the process of the infinite generation of number*s*. Thus we may have an *array* of *numbers* 1, 2, 3, . . . n, all of which are finite. The *semantic process* of *passing* from n to $n+1$ *is not* a number, but constitutes a characteristic of the semantic process. The *result* of the semantic process; namely, $n+1$, again becomes a finite number. If we take a fraction, a/n, the greater an n is selected, the smaller the fraction becomes, but with each selection the fraction again is finite, no matter how small.

Although the two processes are closely connected on the formal side, they are very different from the semantic point of view. The process of generating number*s* may be carried on indefinitely or 'infinitely' and has no upper limit, and we cannot assign such a limit without becoming tangled up in self-contradiction in terms. Not so with the process of indefinite or infinite divisibility. In this case, we start with a *finite*.

Existing mathematical symbolism and formalism lead to identification of both fundamentally different semantic processes and introduce a great deal of avoidable confusion. A $\bar{A}$ orientation will allow us to retain mathematical symbolism and formalism, but will not allow the identification of the semantic process of *passing* from number to number, which passing *is not* a number, with the *result* of this process which, in each case, becomes a *definite and finite number*.

It becomes obvious that the A terminology and present standard notions of 'number' identify the semantic *process* with its *result,* an identification which must ultimately be disastrous. The semantic process is thus potentially infinite, but the *passing* from n to $n+1$ characterizes the semantic process, not number; number*s* representing only finite results of the indefinitely extended semantic process.

A $\bar{A}$ analysis without identification discloses, then, that only the semantic process can be indefinitely extended, but that the *results* of this process, or a number in *each case,* must be *finite*. To speak about an 'infinite' or, as it is called, 'transfinite' 'number', is to identify entirely different issues, and involves very definite self-contradictions in *m.o* terms. The existing mathematical terminology has been developed without the realization of $\bar{A}$ issues and the multiordinality of terms and leads automatically to such identifications. As long as mathematicians do not consider $\bar{A}$ issues, the problems of mathematical infinity will remain unsolved and hopeless; and yet, without a scientific theory of infinity, all of mathematics and most of science would be entirely impossible. A $\bar{A}$ clarification of these problems involves a new semantic definition of number*s* and mathematics, given in Chapter XVIII, which eliminates a great many mysteries in connection with mathematics and does not allow these dangerous and befogging identifications.

From a $\bar{A}$ point of view, we must treat infinity in the first cantorian sense; namely, as a *variable finite,* the term *variable* pertaining to the semantic process but *not* to number, the term *finite* pertaining to both the semantic *arrest* of the infinite semantic process, and so characterizing also its result; namely—a number.

In the meantime, the numerical technique is *indefinitely flexible* in the sense that no matter how great a number we take, we always can, by a semantic process, produce a greater number, and no matter how small the difference between two numbers might be, we always can find a third number which will be greater than the smaller, and smaller than the given greater number. Thus, we see that the *numerical technique* is such as to correspond in flexibility exactly to the *semantic processes,* but

there is nothing flexible about a definite number once it is selected. What has been already said about a variable applies, also, to a number; namely, that a 'variable' does not 'vary' in the ordinary sense; but this term applies only to the semantic processes of the mathematician. The older intensional A definition of 'number' must have led to the older identifications. The $\bar{A}$, extensional, and *non-el* semantic definition of number*s* does not allow such identifications. The A term 'number' applied to a definite number, but also to an intensional definition of number*s*. The $\bar{A}$, or semantic definition of number*s*, is different in the sense that it finds extensional characteristics of each number, applicable to all number*s*, and so helps not to identify a definite number with the process of generating number*s*, which the use of one term for two entirely different entities must involve.

Cantorian *alephs,* then, are the result of identification or confusion of entirely different issues and must be completely eliminated. The rejection of *alephs* will require a fundamental revision of those branches of mathematics and physics which utilize them; yet, as far as I know, with a very few exceptions, the *alephs* are not utilized or needed, although the *'name'* is used, which spell-mark has become fashionable in many mathematical and physical circles. In the case of *alephs,* history may repeat itself and the *alephs,* like the 'infinitesimal', when their self-contradictory character becomes understood, will be eliminated without affecting the great body of mathematics, but only the small portions which are built on the *alephs.*

As to the existence of infinite processes, we know positively *only* about the *semantic process* of generating number*s* and the *semantic process* of infinite divisibility. These processes are evident in our common experience. We cannot *a priori* know if such infinite processes can be found in the world which must be discovered by investigation and experimentation.

The existing terminology is still A and is based on, and leads to, identification, and so in my $\bar{A}$ presentation I cannot use it and expect to clear up some of the issues involved. The terms such as 'class', 'aggregate', 'set'., imply a definite static collection. The term 'infinite', in the meantime, can only be correctly and significantly used as applied to a dynamic semantic process. We cannot speak of 'infinite' classes, aggregates, sets., and evade the issues of identification of entirely different entities. The term 'series' has a technical meaning in connection with numbers and so, for a general discussion of processes, is a little too specific. The term 'array' is more general, yet extensional, of which 'series' would be a special case. The general term 'number'

is multiordinal and intensional and so, in the $\bar{A}$ extensional system, ∞-valued, and must be used in the plural; namely, 'number*s*'. The term 'number' in the singular will be used to indicate a definite number. The term 'denumerable' has been introduced by Cantor and means any extensional array of terms, facts, states, observables., which can be put in one-to-one correspondence with the infinite array of positive integers.

Let me repeat once more: the semantic process may be carried on without limits, and the infinite series of positive integers is an extensional, technical, and verbal expression of this semantic process and the only infinite array of which existence we are certain.

We shall be able to explain, and to give a better definition of, mathematical infinity if we introduce an extremely useful structural term, 'equivalence'. Two processes, arrays., between which it is possible to set up, by some law of transformation, a *one-to-one* correspondence are said to be *equivalent*. A process, array., which is equivalent to a part of itself, is said to be infinite. In other words, a process, array., which can be put into a *one-to-one* correspondence with a part of itself is said to be infinite. We can define a *finite* process, array., (class, aggregate.,) as one which is not infinite. The following is valid *exclusively* because of the use of the '**etc.**'

A few examples will make this definition clearer. If we take the series of positive integers, 1, 2, 3, 4, . . . **etc.**, we can always double every number of this row *provided* we retain the *process-character, but not otherwise*. Let us write the corresponding row of their doubles under the row of positive integers, thus:

1, 2, 3, 4, 5, ... **etc.**
2, 4, 6, 8, 10, . . . **etc.**

or we can treble them, or *n*-ble them, thus:

1, 2, 3, 4, 5, ... **etc.**
3, 6, 9, 12, 15, . . **etc.**

there are obviously as many numbers in each row below as in the row above, *provided we retain the* '**etc.**', so the number*s* of number*s* in the two rows compared must be equal. All numbers which appear in each bottom row also occur in the corresponding upper row, although they only represent a *part* of the top row, *again provided that we retain the* '**etc.**'

The above examples show another characteristic of infinite processes, arrays,. In the first example, we have a *one-to-one* correspondence between the natural number*s* and the *even* number*s*, which are equal in number at each stage. Yet, the second row results from the first row by

taking away all odd numbers, which, itself, represents infinite numbers of numbers.

This example was used by Leibnitz to prove that infinite arrays cannot exist, a conclusion which is not correct, since he did not realize that both finite and infinite arrays depend on definitions. We should be careful not to approach *infinite* processes, arrays., with prejudices, or silent doctrines and assumptions, or, in general *s.r*, taken over from *finite* processes, arrays,.

Thus we see that the process of generating natural numbers is structurally an infinite process because its *results* can be put in a *one-to-one* correspondence with the results of the process of generating even numbers., which is only a part of itself. Similarly, a line AB has infinitely many points, since its points can be put into a *one-to-one* correspondence with the points on a segment CD of AB. Another example can be given in the Tristram Shandy paradox of Russell. Tristram Shandy was writing his autobiography, and was using one year to write the history of one day. The question is, would Shandy ever complete his biography? He would, provided he never died, or he lived infinite numbers of years. The hundredth day would be written in the hundredth year, the thousandth in the thousandth year, **etc.** No day of his life would remain unwritten, *again provided his process of living and writing would never stop.*

Such examples could be given endlessly. It is desirable to give one more example which throws some light on the problems of 'probability', 'chance',. The theory of probability originated through consideration of games of chance. Lately it has become an extremely important branch of mathematical knowledge, with fundamental structural application in physics, general semantics, and other branches of science. For instance, Boltzman based the second law of thermodynamics on considerations of probability. Boole's 'laws of thought', and the many-valued 'logic' of Łukasiewicz and Tarski are also closely related to probability; and the new quantum mechanics uses it constantly,.

The term 'probability' may be defined in the rough as follows: If an event can happen in a different ways, and fails to happen in b different ways, and all these ways are equally likely to occur, the probability of the happening of the event is $\frac{a}{a+b}$, and the probability of its failing is $\frac{b}{a+b}$.

Let us assume that in a certain city a lecture is held each day, and that, though the listeners may change each day, the numbers of listeners

are always equal. Suppose that one in each twenty inhabitants of this town has M as the first letter of his name. What is the probability that, *'by chance'*, all the names of the audience would begin with M? Let us call such a happening the M-event. In the simplest case, when the daily number of listeners is only one, the probability of an M-event is 1 in 20, or 1/20. The probability of an M-event for an audience of 2 is 1 in $20 \times 20 = 400$, or 1/400. The probability that an audience of three members should have all three names begin with M would decrease twenty times further. Only once in 8000 lectures, on an average, would an M-event happen. For five people it would amount to 1 in $20 \times 20 \times 20 \times 20 \times 20 = 3{,}200{,}000$ days, or 1/3,200,000, or once in approximately 9000 *years;* for ten people, about once in thirty billion years; for twenty people, about once in a third of a quadrillion years. For one hundred people, the recurrence period of the M-event would be given as once in a number of years represented by more than a hundred figures. If the town, in this last example, should be as old as the solar system, and if the lectures had been delivered daily to an audience of one hundred people through this inconceivably long period, the probability is extremely small that the M-event would happen at all.[2]

From the human, *anthropomorphic,* point of view, we would say that such an event is impossible. But it must be remembered that this is only an anthropomorphic point of view, and our judgements are coloured by the temporal scale of our own lives. Of course, to carry such an anthropomorphic viewpoint into cosmic speculations is simply silly, a survival of the primitive structure of language and its progeny—metaphysics and mythologies.

The theory of infinity throws considerable structural light on such primitive speculations. In this external world, we deal with processes, and, as we measure 'length' by comparison with freely selected convenient units of 'length', let us say, an inch; or we measure 'volume' by freely selected convenient units of 'volume'; so, also, we compare *processes* with some freely selected and convenient *unit-process*. The diurnal rotation of our earth is such a process, and, if we choose, we can use it as a measuring unit or as a comparison standard. Of late, we have become aware that the rotation of the earth is not quite regular, and so, for accurate measurements, the old accepted unit-process of a day, or its subdivision, a second, is not entirely satisfactory. For scientific purposes, we are trying to find some better unit-process, but we have difficulty, as the problem is naturally circular. When we speak in terms of a 'number of years', or of seconds, we speak about perfectly good observational experimental facts, about quite definite relations, the best we know in

1933. We do not make any metaphysical assertions about 'time' and we should not be surprised to find that statements involving 'years' are generally propositions, but that statements involving 'time' often are not. It is necessary not to forget this to appreciate fully what follows.

The theory of infinity will clear away a troublesome stumbling-block. We will use the expression 'infinite number*s* of years', remembering the definition of 'infinite number*s*' and what was said about the *unit-process* which we call a year. We have seen in an example above that if only a hundred individuals attend a lecture, and all 'by chance' have their names begin with M, such an event happens, on an average, only once in an inconceivably large number of years, represented by a number with a hundred figures. If we would ask *how many* times an occurrence would happen, we would have to state the period in years for which we ask the *how many*. It is easy to see that in infinite number*s* of years, this humanly extremely rare occurrence would happen precisely *infinite numbers of times,* or, in other words, 'just as often', this last statement being from a non-anthropomorphic point of view. An event that appears, from our human, limited, anthropomorphic point of view, as 'rare', or as 'chance', when transposed from the level of finite process, arrays . , to that of infinite processes, arrays . , is as 'regular', as much a 'law', involving 'order', as anything else. It is the old primitive *s.r* to suppose that man is the only measure of things.

Here the reader might say that infinite number*s* of years is a rather large assumption to be accepted so easily. This objection is indeed serious, but a method which can dispose of it is given later on. At this stage, it is sufficient to say that, on the one hand, this problem is connected with the semantic disturbance, called identification (objectification of 'time'), which afflicts the majority of us, excepting a few younger einsteinists ; and that, on the other hand, it involves the structurally reformulated law of the 'conservation of energy', 'entropy', .

Before parting with the problem of infinity, let me say a word about the notion of 'continuity', which is fundamental in mathematics. Mathematical continuity is a structural characteristic connected with ordered series. The difficulties originated in the fact that a 'continuous' series must have infinite number*s* of terms between any two terms. Accordingly, these difficulties are concerned with infinity. That mathematicians need some kind of continuity is evident from the example of two intersecting lines. If the lines have gaps, as, for instance, — — — — , there would be the possibility that two gaps would coincide, and the two lines not intersect ; although in a plane the first line would pass to the other side of the second line. At present, we have two kinds of 'con-

tinuity' used in mathematics. One is a supposedly 'high-grade' continuity; the other, supposedly, is a 'low-grade' continuity, which is called 'compactness' or 'density', with the eventual possibility of gaps. I am purposely using rather vague language, since these fundamental notions are now being revised, with the probability that we shall have to be satisfied with 'dense' or 'compact' series and abandon the older, perhaps delusional, 'high-grade' continuity. It is interesting to note that the differential and integral calculus is supposedly based on the 'high-grade' continuity, but the calculus will not be altered if we accept the 'low-grade' compactness, all of which is a question of an A or $\bar{A}$ orientation.

Vague feelings of 'infinity' have pervaded human *s.r* as far back as records go. Structurally, this is quite natural because the term infinity expresses primarily a most important semantic process. The majority of our statements can also be reformulated in a language which explicitly involves the term 'infinity'. An example has been already given when we were speaking about the universal propositions which were supposed to be of *permanent* validity, in other *language,* valid for 'infinite numbers of years'. We see how the trick is done—a vague quasi-qualitative expression like 'permanent' or 'universal' is translated into a quantitative language in terms of 'numbers of years'. Such translation of qualitative language into quantitative language is very useful, since it allows us to make more precise and definite the vague, primitive structural assumptions, which present enormous semantic difficulties. This brings to our attention more clearly the structural facts they supposedly state, and aids analysis and revision. In many instances, such translations make obvious the illegitimacy of the assumptions of 'infinite velocities' and so clear away befogging misunderstandings, and beneficially affect our *s.r.*

CHAPTER XV

THE 'INFINITESIMAL' AND 'CAUSE AND EFFECT'

> But we are not likely to find science returning to the crude form of causality believed in by Fijians and philosophers, of which the type is "lightning causes thunder." (457) BERTRAND RUSSELL
>
> The notion of causality has been greatly modified by the substitution of space-time for space and time. . . . Thus geometry and causation becomes inextricably intertwined. (457) BERTRAND RUSSELL

> In classical mechanics, and no less in the special theory of relativity, there is an inherent epistemological defect which was, perhaps for the first time, clearly pointed out by Ernst Mach. . . . No answer can be admitted as epistemologically satisfactory, unless the reason given is an *observable fact of experience*. The law of causality has not the significance of a statement as to the world of experience, except when *observable facts* ultimately appear as causes and effects. (155) A. EINSTEIN

> The chain of cause and effect could be quantitatively verified only if the whole universe were considered as a single system—but then physics has vanished, and only a mathematical scheme remains. The partition of the world into observing and observed system prevents a sharp formulation of the law of cause and effect. (215) W. HEISENBERG

Of late, another perplexing semantic problem concerning 'causality' or 'non-causality' has arisen in connection with the newer quantum mechanics. It is possible to examine this question by different methods. The simpler one is connected with vague feelings of 'infinity' and its supposed opposite, the 'infinitesimal'; the more fundamental method is based on the orders of abstractions leading toward the ∞-valued semantics of probability.

Because of man's natural tendency to speak in terms of 'infinity', and his further marked tendency of having opposites, such as 'yes', 'no', 'right', 'left', 'positive', 'negative', 'love', hate', 'honesty', 'dishonesty'., quite naturally the notion of 'infinity' carried with it the tendency to invent the 'infinitesimal'. Even mathematicians have had great semantic difficulties in breaking away from this habit. Analysis persistently reveals that structurally no matter how far we go in dividing something, let us say an inch, whatever is left may be extremely small, but yet it is a perfectly good *finite* quantity. Thus, structural difficulties were encountered with the postulated 'infinitesimal'. The *name* implies that they are not finite, yet analysis shows only finites. Mathematicians supposed that an 'infinitesimal' was necessary for mathematics, and so they were reluctant to abandon it.

The 'infinitesimal', like so many other baffling suppositions, was invented by the Greeks, who regarded a circle as differing 'infinitesimally'

from a polygon with a very large number of very small equal sides. With the invention of the differential and integral calculus, 'infinitesimal calculus', as it was called, the importance of the 'infinitesimal' increased, and even mathematicians used it as a fundamental notion. Finally, Weierstrass succeeded in showing the meaningless character of the 'infinitesimal', and also that the 'infinitesimal' was not structurally necessary for the calculus. Up to that date, the problem was baffling; we knew that the calculus required 'continuity', which, in turn, seemed to require 'the infinitely little', and yet no one could tell what this 'infinitely little' might represent. It was quite obviously not zero, because a sufficient number of them was able to make up a finite whole; and we knew no fraction which was not zero and yet not finite. The discovery by Weierstrass that the calculus does not require the 'infinitesimal', and that all deductions could be made without it, abolished a very serious structural, verbal, metaphysical, and semantic bugaboo. Common sense, of course, is much simpler, although unreliable in such matters, and was satisfied also.

The elimination of the 'infinitesimal' is a great semantic step forward, and helps to clarify structurally some deeply rooted, vague, fallacious notions, which are overloaded with affective components and are extremely vicious in their effects.

If there is no 'infinitesimal', there is no 'next moment'; for the interval between any two moments must be finite, and so there are always other moments in the interval between them. Also, two moments cannot be consecutive, for between any two there are always other moments, no matter how far we go; similarly, the 'present' becomes a very vague notion.

For our purpose, the most fundamental semantic application of what has been said above is in the vast field embraced by the old structural notions of 'cause' and 'effect'. These terms are of great antiquity, of a distinctly pre-scientific one-, two-valued semantic epoch. They originated in the rough experience of our race, and are firmly rooted in the habits of 'thought' and the structure of our old two-valued 'logic' and language, and because of that are even now unduly baffling. These terms, in the *two-valued sense,* were and are the structural assumptions of our 'private' and 'official' 'philosophies'. The unenlightened use of these terms has done much to prevent the formulation of a science of man and to build up vicious anti-scientific metaphysics of various sorts involving pathological *s.r.* With the new quantum mechanics, a better understanding of these notions, based on the ∞-valued semantics of probability, becomes a paramount issue for all science. In daily life, the

indiscriminate use of two-valued 'cause' and 'effect' leads structurally to a great deal of absolutism, dogmatism, and other harmful semantic disturbances, which I call confusion of orders of abstraction.

We usually follow the 'philosophers' and ascribe—or, rather feel, as conscious ascribing would not stand criticism—some mysterious structural continuity, some mysterious overlapping of 'cause' and 'effect'. We 'feel', and try to 'think', about 'cause and effect' as *contiguous* in 'time'. But 'contiguous in time' involves the impossible 'infinitesimal' of some unit of 'time'. But, since we have seen that there is no such thing, we must accept that the interval between 'cause' and 'effect' is finite. This structural fact changes the whole situation. If the interval between 'cause' and 'effect' is finite, then always something might happen between, no matter how small the interval may be. The 'same cause' would not produce the 'same effect'. The expected result would not follow. This means only that in this world, to be sure of some expected effect, requires that there must be nothing in the environment which can interfere with the process of passing from the conditions labelled 'cause' to the conditions labelled 'effect'. In this world, with the structure which it has, we can never suppose that *a* 'cause', as we know it, is *alone* sufficient to produce the supposed 'effect'. When we consider the ever-changing environment, the number of possibilities increases enormously. If it were possible to take into account the *whole* of the environment, the *probability* that some event would be repeated, in all details, thus exhibiting the assumed two-valued relation of 'cause' and 'effect', which we took for granted in the old days, would practically be nil. The principle of non-elementalism, as we see, requires an ∞-valued semantics of probability.

The reader should not take what is said here as a denial that in this external world some regularities of sequence occur; but the above analysis, which is mainly due to Russell,[1] shows clearly that the verbal principle of 'same cause, same effect' is structurally untenable. We can never manage to observe the 'same cause' in detail. As soon as the antecedents have been sufficiently ascertained, so as to calculate the consequences with some plausible accuracy as to details, the relations of these antecedents have become so complex that there is very little *probability* that they will ever occur again.

The clearing up of the problems of 'cause' and 'effect' is of serious importance, because powerful semantic reactions are connected with it. To begin with, we must differentiate between the terms 'cause' and 'effect', which, linked together, imply a two-term relation nowhere to be found in this world, and thus represent a language and a two-valued

'logic' of a structure not similar to the structure of the world, and the *general ∞-valued notion of causality*. This last notion is the psychological foundation of all explanations leading toward ∞-valued determinism, and is an exclusive test for structure; and so of extreme semantic importance.

Besides the analysis from the point of view of the impossible 'infinitesimal', the term 'cause-effect' represents a two-term relation, and, as such, is a primitive generalization *never* to be found in this world, as all events are *serially* related in a most complex way, independent of our way of speaking about them. If we expand our two-term relation 'cause-effect' into a *series,* we pass from the inferential level to the *descriptive* level, and so can apply a behaviouristic, functional, actional language of order. In such series, we could only use the language of 'cause' and 'effect' if we could select neighbouring factors, a selection which is often impossible. Also, if we pass from macroscopic to microscopic or sub-microscopic levels, we could use such language, but then the terms would have different meanings, supplied by the theory of probability.

The semantic side of this problem is of importance, because, in the old *el* way, it was neglected. General speculations about such *m.o* terms as 'cause' and 'effect' are useless. Such statements are not propositions, but involve variable meanings and, therefore, generate propositional functions which are neither true nor false. Our expanding of the too simple, two-term relation 'cause-effect' into a complex series is closer to the structure of this world, as far as we know it.

The understanding and habitual application of what has just been said would not only save us from silly dogmatizing and inappropriate *s.r*, but would teach us not to disregard any regularity, and to investigate any relation which might appear. Then, in a *specific case,* we could again use the restricted principle of causality, based on probability and averages. The old absolute and objectified semantic attitude toward 'cause-effect' was and often is a serious hindrance in observing impartially the sequence of events (order) and relations. Preconceived notions and old *s.r* played havoc, for it is well known that we usually find what we want to find. If we approach a problem with definite unconscious *'emotional'* wants, and cannot satisfy these *s.r*, we become bewildered, down-hearted, and perhaps utter some such non-sense as the 'finite mind', or the like. Under such semantic pressure, our power of observation and analysis is reduced by a kind of 'emotional stupor'. Such an occurrence is harmful in science and in life. 'Human knowledge' depends on human ingenuity, power of observation, power of abstraction , . It is an activity

of the human nervous system inside of our skin and can never be the events themselves.

We see that the old two-valued verbal structure of 'cause' and 'effect' is not similar to the structure of the world, but a rash limiting generalization from probability. Since these expressions belong to the class of statistical averages and depend on the scale of the events and intervals dealt with, we must not expect that such terms as two-valued 'causality', which is a term of statistical macroscopic averages, will apply in that sense to small-scale events when the intervals are much smaller and when entirely different conditions and 'causes' prevail. Today we have structural evidence that even 'space' and 'time' represent statistical averages and do not apply to the smallest scale events. It is natural that 'cause' and 'effect' should join their company. The above involves epistemologically the passing from the A two-valued system to a $\bar{A}$ ∞-valued system. Psychophysiologically, it involves new *s.r.*

In mathematics, the old religious attitude toward the 'infinitesimal' is rapidly vanishing. Many mathematicians deliberately, and justly, avoid the use of the word. A term like 'indefinitely small' or 'indefinitesimal' is a better descriptive term, truer in its implications. We even see scientists like Eddington, who had the pluck—it is still pluck, unfortunately—to treat enormous stellar distances as 'infinitesimals of second order'. ('Infinitesimal' is used here in a mathematical sense of indefinitesimal.)

It has been already mentioned that most of the important discoveries of mathematics were due to a special semantic attitude on the part of those who made them. This attitude was an unconscious or conscious treatment of mathematics as a form of human behaviour. We see an example in the work of Weierstrass and his analysis of the 'infinitesimal'. He did not take the 'infinitesimal' as some objectified metaphysical structure and remain content; he analysed the *genetic process* by which the 'infinitesimal' was *made* by Smith and Brown, and so treated mathematics structurally as a form of human behaviour. Any deepening of the foundations, or clarification of fundamental notions, or investigation of underlying assumptions . , *must,* by necessity, have this characteristic. The man who does it must take into account how the given process was produced—analyse its structure, and so start with the ways and methods of production. In other words, he must treat the given problem as a form of human behaviour. The fact that this simple and quite obvious method has been formulated and structurally explained as *desirable* is helpful. It shows the *method* and structure of the path by which advances can be reached. We can *train* the semantic reactions of students to it and make progress inevitable; but now, instead, it takes a genius to break, by him-

self, through the old semantic habits which have been produced by the lack of scientific psycho-logics and training.

The term 'correct symbolism' has already been used. In this world of structurally absolute individuals, the minimum of structurally desirable correct symbolism must provide for the possibility of labelling these absolute individuals by separate names. For scientific purposes, we must use terms built on the pattern of mathematical symbolism; i.e., according to the *extensional* methods. We must adopt a behaviouristic attitude and habits in our term-making. As we proceed, we must emphasize *order,* considering what comes first and what next. This is semantically important, for the usual procedure is entirely different: first, we have our structurally 'preconceived' doctrines and languages; next, we observe the structure of the world; and *then* we try to force the observed facts into the linguistic structural patterns. But, in the new way, we *start* with silent observations, and search empirically for structure; next, we invent verbal structures similar to them; and, finally, we see what can be said about the situation, and so test the language. Experience shows that the old habits of labels first, objects next, instead of the structurally natural order of *objects first, labels next,* is semantically pernicious and harmful. In Part VII, it is shown that the semantic structural reversal of the unnatural reversed order is crucial for sanity.

From the days of the Greeks an acute difficulty has made itself felt; namely, how to reconcile the world of physics with the world of mathematics. For mathematics, we need 'extensionless' points; for physics, we need finite-sized elements. Whitehead and Russell have suggested different structures by which this may be accomplished. It seems possible to demand that none of the material dealt with shall be smaller than an assigned finite size. That this condition can be reconciled with mathematical continuity seems to be novel. Whether this device is valid or not, it is yet too early to decide. This problem of reconciliation will become important further on when we come to speak of events as made up from point-events.[2]